MAT2002 **Name (Print):** _____
Spring 2026
HW5
Deadline: 26th April

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1. Find the general solution of the given system of equations: $\mathbf{x}' = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ -8 & -5 & -3 \end{pmatrix} \mathbf{x}$

2. Solve the given initial value problem. Describe the behavior of the solution as $t \to \infty$:

   (i) $\mathbf{x}' = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 2 & 2 \\ -1 & 1 & 3 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$

   (ii) $\mathbf{x}' = \begin{pmatrix} 1 & -5 \\ 1 & -3 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \binom{1}{1}$

3. Express the general solution of the given system of equations in terms of real-valued functions.

$$\mathbf{x}' = \begin{pmatrix} -3 & 0 & 2 \\ 1 & -1 & 0 \\ -2 & -1 & 0 \end{pmatrix} \mathbf{x}$$

4. Find a fundamental matrix for the given system of equations and also find the fundamental matrix $\mathbf{\Phi}(t)$ satisfying $\mathbf{\Phi}(0) = \mathbf{I}$:

$$\mathbf{x}' = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ -8 & -5 & -3 \end{pmatrix} \mathbf{x}$$

5. Consider an oscillator satisfying the initial value problem

$$u'' + \omega^2 u = 0, \quad u(0) = u_0, \quad u'(0) = v_0$$

   (i) Let $x_1 = u, x_2 = u'$, and transform the above IVP into the form

$$\mathbf{x}' = \mathbf{A}\mathbf{x}, \quad \mathbf{x}(0) = \mathbf{x}^0.$$

   (ii) By using the Taylor series of exp, show that

$$\exp(\mathbf{A}t) = \mathbf{I}\cos\omega t + \mathbf{A}\frac{\sin\omega t}{\omega}.$$

   (iii) Find the solution of the initial value problem in (i).

6. Solve $\mathbf{x}' = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 1 & -1 \\ 0 & -1 & 1 \end{pmatrix} \mathbf{x}$

7. Solve the IVP:$\mathbf{x}' = \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 3 & 6 & 2 \end{pmatrix} \mathbf{x}, \quad \mathbf{x}(0) = \begin{pmatrix} -1 \\ 2 \\ -30 \end{pmatrix}$

8. Let

$$\mathbf{J} = \begin{pmatrix} \lambda & 1 & 0 \\ 0 & \lambda & 1 \\ 0 & 0 & \lambda \end{pmatrix},$$

where λ is an arbitrary real number.

(i) Find $\mathbf{J}^2, \mathbf{J}^3$, and $\mathbf{J}^4$.

(ii) Use an inductive argument to show that

$$\mathbf{J}^n = \begin{pmatrix} \lambda^n & n\lambda^{n-1} & [n(n-1)/2]\lambda^{n-2} \\ 0 & \lambda^n & n\lambda^{n-1} \\ 0 & 0 & \lambda^n \end{pmatrix}$$

(iii) Determine $\exp(\mathbf{J}t)$.

9. In each of following, find the general solution of the given system of equations.

(i) $\mathbf{x}' = \begin{pmatrix} 2 & -1 \\ 3 & -2 \end{pmatrix} \mathbf{x} + \binom{e^t}{t}$

(ii) $\mathbf{x}' = \begin{pmatrix} 1 & \sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix} \mathbf{x} + \binom{e^t}{\sqrt{3}e^{-t}}$

(iii) $\mathbf{x}' = \begin{pmatrix} 2 & -5 \\ 1 & -2 \end{pmatrix} \mathbf{x} + \binom{\csc t}{\sec t}, \quad \frac{\pi}{2} < t < \pi$